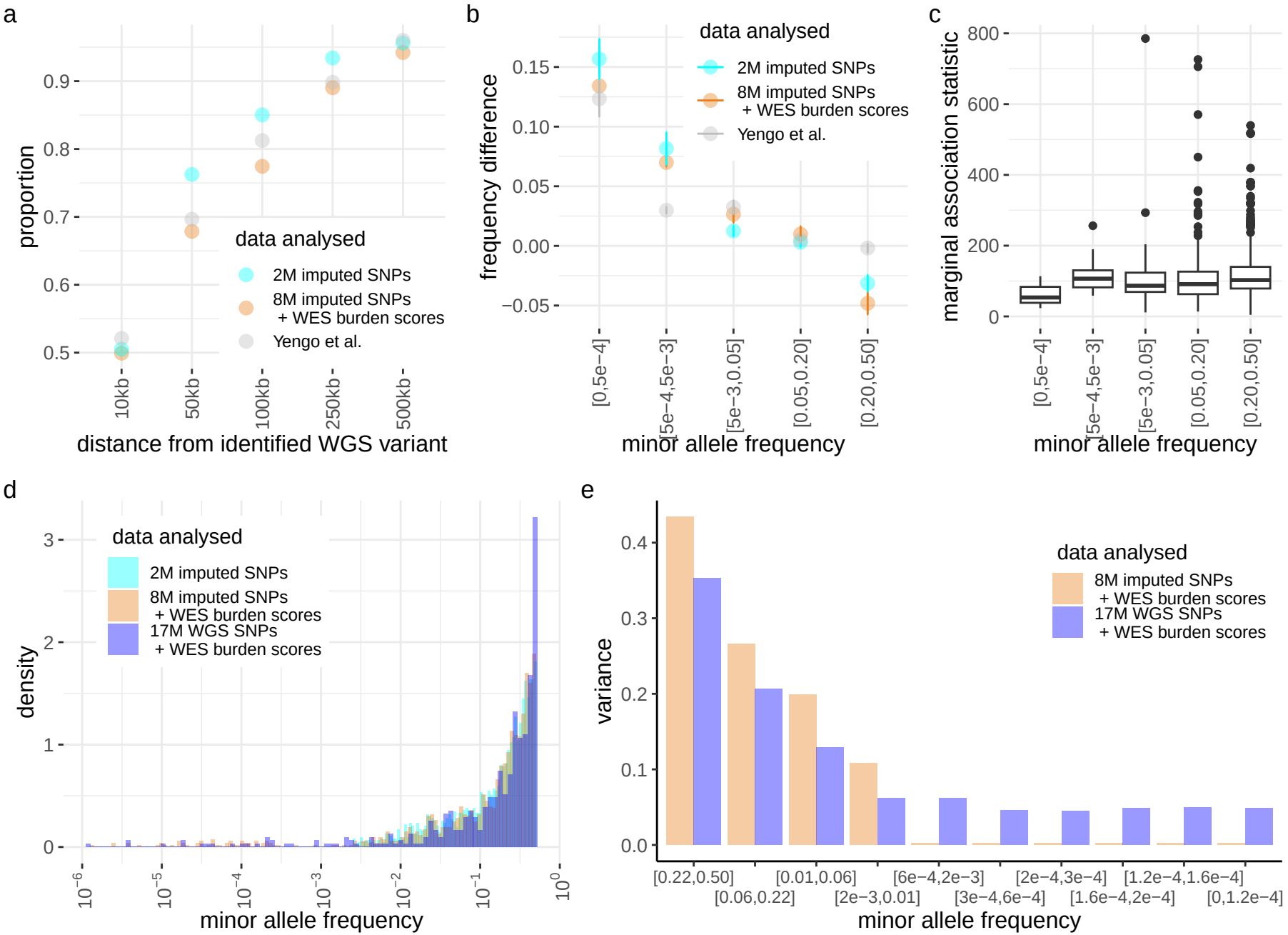
a
proportion
0.9
0.8
0.7
0.6
0.5
data analysed
2M imputed SNPs
8M imputed SNPs
+ WES burden scores
Yengo et al.
10kb
50kb
100kb
250kb
500kb
distance from identified WGS variant
b
frequency difference
0.15
0.10
0.05
0.00
−0.05
data analysed
2M imputed SNPs
8M imputed SNPs
+ WES burden scores
Yengo et al.
[0,5e−4]
[5e−4,5e−3]
[5e−3,0.05]
[0.05,0.20]
[0.20,0.50]
minor allele frequency
c
marginal association statistic
800
600
400
200
0
[0,5e−4]
[5e−4,5e−3]
[5e−3,0.05]
[0.05,0.20]
[0.20,0.50]
minor allele frequency
d
density
3
2
1
0
data analysed
2M imputed SNPs
8M imputed SNPs
+ WES burden scores
17M WGS SNPs
+ WES burden scores
10^−6
10^−5
10^−4
10^−3
10^−2
10^−1
10^0
minor allele frequency
e
variance
0.4
0.3
0.2
0.1
0.0
data analysed
8M imputed SNPs
+ WES burden scores
17M WGS SNPs
+ WES burden scores
[0.22,0.50]
[0.06,0.22]
[0.01,0.06]
[2e−3,0.01]
[6e−4,2e−3]
[3e−4,6e−4]
[2e−4,3e−4]
[1.6e−4,2e−4]
[1.2e−4,1.6e−4]
[0,1.2e−4]
minor allele frequency